

Student Attendance and Performance Project

CSE1021 – Introduction to Problem Solving and Programming

1. Introduction

I made this project to practise the Python topics we learned in class. It is a simple console program for keeping student details, attendance and marks together.

2. Problem

When student details, attendance and marks are kept separately, it can take extra time to find a complete record. I wanted to make a small program that keeps these details together and gives a quick summary.

3. Main Features

- Add, list, search and delete students.
- Record attendance and calculate the percentage.
- Enter marks for different subjects.
- Calculate average marks and a grade.
- Show a student summary.
- Save information in a local JSON file.

4. How the Program Works

The program starts with a menu. The user selects an option, enters the required information, and the program checks the input. The data is then saved or the requested result is displayed. The menu appears again until the user chooses Exit.

5. Implementation

Most of the working code is in **main.py** because this is a small project. **storage.py** handles the JSON file. **algorithms.py** contains basic calculation functions, and **test_project.py** contains a few tests.

6. Calculations

Attendance percentage = (classes attended / total classes) × 100. The average mark is calculated by adding the subject marks and dividing by the number of subjects. The grade is selected from the ranges used in the program.

7. Testing

I included simple tests for attendance calculation, average marks, the data structure and the grade range. The current test run passes all 4 tests.

8. Limitations

This is a basic academic project. It does not have a GUI, login system or online database. The records are stored only in the local JSON file.

9. Future Improvements

If I continue the project, I would like to add a GUI, a database and an option to export reports.

10. Learning Outcome

While making this project, I practised functions, conditions, loops, dictionaries, file handling, input validation and basic testing.